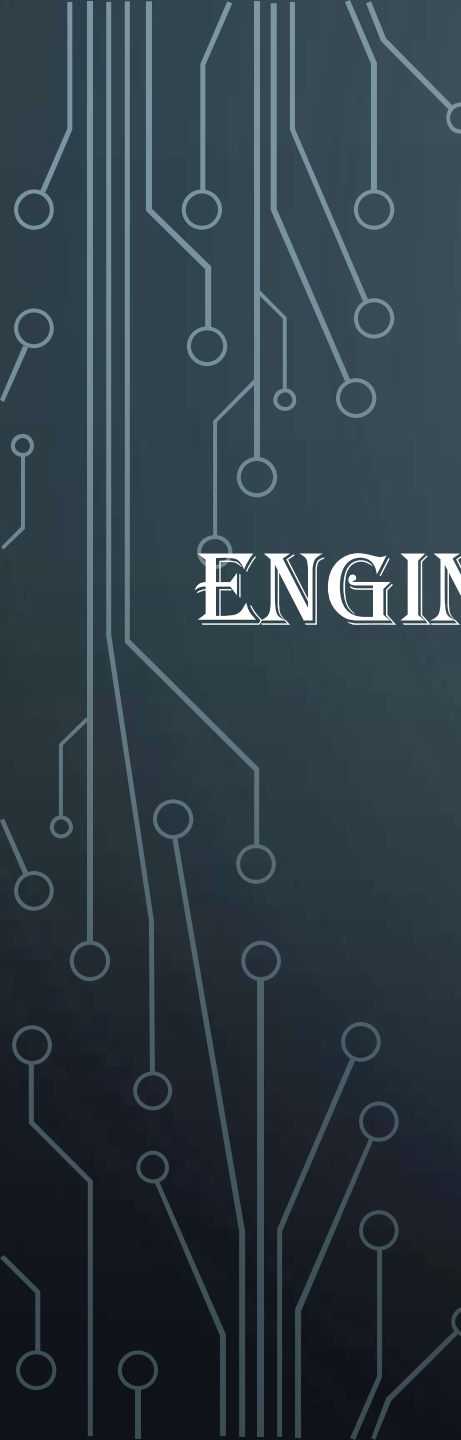




ENGINEERING EXPLORATION FOR COMPUTER SCIENCE AND ENGINEERING

WHAT IS ENGINEERING?

Engineering is the use of scientific knowledge to design, create, and improve machines, buildings, and technology.



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING.

Science

Studies natural things.

Gains knowledge through experiments.

Explains facts.

Engineering

Uses scientific knowledge.

Creates useful products.

Solves human problems



DIFFERENCE BETWEEN SCIENTIST AND ENGINEER.

Scientist

Studies nature.

Performs research.

Discovers new ideas.

Engineer

Applies scientific knowledge.

Designs and builds products.

Improves technology.



EXPLAIN THE NEEDS AND WANTS OF TODAY'S ENGINEERS (OR MODERN ENGINEERS).

Technical knowledge

Computer skills

Critical thinking

Communication skills

Teamwork



LIST AND EXPLAIN THE VARIOUS DISCIPLINES OF ENGINEERING. ALSO, MAP THE ROLE OF COMPUTER SCIENCE ENGINEERING WITH OTHER BRANCHES OF ENGINEERING USING REAL-WORLD PRODUCTS AS CASE STUDIES.

Engineering Disciplines:

Computer Science Engineering (CSE) – Software and computing.

Mechanical Engineering – Machines and manufacturing.

Civil Engineering – Buildings and infrastructure.

Electrical Engineering – Power systems.

Electronics & Communication Engineering (ECE) – Electronics and communication. .

Role of CSE with Other Branches:

Mechanical – Robotics and automation.

Civil – Smart city and traffic management software.

Electrical – Smart grids and power monitoring.

ECE – Smartphones and embedded systems.

Biomedical – Medical imaging and health monitorin



WRITE SHORT NOTES ON THE MISCONCEPTIONS ABOUT COMPUTER SCIENCE (CS) ENGINEERING.

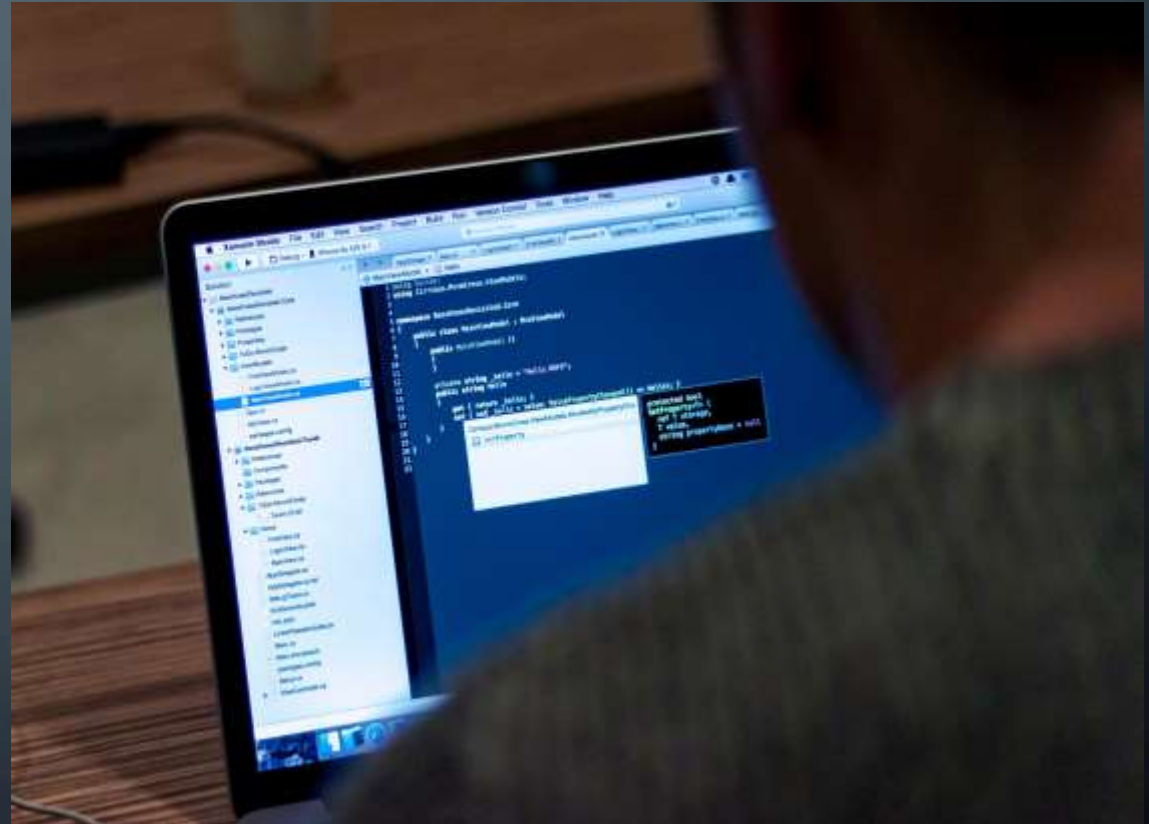
CSE is only about coding.

Engineers only repair computers.

Mathematics alone is enough.

CSE jobs are only in IT companies.

AI will replace all programmers.



WHAT SKILLS ARE REQUIRED FOR A COMPUTER SCIENCE (CS) ENGINEERING STUDENT?

Programming skills

Problem-solving ability

Logical thinking

Communication skills

Teamwork

Creativity

Knowledge of databases and
networks



The background is a dark blue gradient. In the corners, there are white line art illustrations of circuit boards or neural networks, with lines connecting to small circles.

THANK YOU

BY MUGILAN